

B3 Atomic, Quantum and Laser Physics, 2025 — Model Solutions

Question 1 — Two-electron atoms: Ca and Ba

Problem. Ca ($Z = 20$) and Ba ($Z = 56$) level diagrams labelled X, Y; closed core + two valence s electrons. Identify, label, and discuss term/level structure and E1 transitions.

(a) Central-field approximation and configurations

Exact Hamiltonian

$$H = \sum_i \left(-\frac{\hbar^2}{2m_e} \nabla_i^2 - \frac{Ze^2}{4\pi\epsilon_0 r_i} \right) + \sum_{i < j} \frac{e^2}{4\pi\epsilon_0 r_{ij}}$$

is non-separable. Replace mean effect of other electrons by a spherical $V_c(r_i)$ interpolating between bare ($-Ze^2/4\pi\epsilon_0 r$) and fully screened ($-e^2/4\pi\epsilon_0 r$). Then

$$H_0 = \sum_i \left[-\frac{\hbar^2}{2m_e} \nabla_i^2 + V_c(r_i) \right] \quad (1)$$

separates; one-electron states carry (n, ℓ, m_ℓ, m_s) with ℓ -dependent energies (low- ℓ orbitals penetrate the core). Filling shells antisymmetrically by Pauli gives the **electron configuration** $(1s)^2(2s)^2 \dots$.

(b) Terms and levels

H_0 -degeneracy of a configuration is lifted by:

- Residual electrostatic H_1 : commutes with $\mathbf{L}, \mathbf{S}$, so splits the configuration into LS **terms** ^{2S+1}L . Exchange part favours high S (Hund's first rule).
- Spin-orbit $H_{SO} = \sum_i \zeta(r_i) \ell_i \cdot \mathbf{s}_i$: couples $\mathbf{L}, \mathbf{S} \rightarrow \mathbf{J}$, splitting each term into **levels** $^{2S+1}L_J$, $J = |L - S|, \dots, L + S$.

Light atoms: term-splitting $\gg$ fine structure (LS coupling). Heavy atoms drift toward jj .

(c) Identification

Both have $(ns)^2$ ground giving 1S_0 ; lowest excited configurations are $ns(n-1)d$ and $nsnp$. In X, $ns(n-1)d$ lies *below* $nsnp$ and the triplet fine structure ($\sim 10^3 \text{ cm}^{-1}$) is large enough to be visible directly; in Y the order is reversed and the fine structure is so small ($\sim 50 \text{ cm}^{-1}$) that a $\times 20$ inset is needed. In heavy atoms $(n-1)d$ is pulled in and fine structure $\sim Z^4 \alpha^2$ is large, so

$X = \text{Ba} (Z = 56), \quad Y = \text{Ca} (Z = 20).$	(2)
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(d) Term and level labels

Closed $(ns)^2$:

$$(ns)^2 : \quad {}^1S_0.$$

$nsnp$ ($\ell = 0, 1 \Rightarrow L = 1; S = 0, 1$):

$$nsnp : \quad {}^3P_0, {}^3P_1, {}^3P_2, {}^1P_1.$$

$ns(n-1)d$ ($\ell = 0, 2 \Rightarrow L = 2; S = 0, 1$):

$$ns(n-1)d : \quad {}^3D_1, {}^3D_2, {}^3D_3, {}^1D_2.$$

Ba: $6s^2$, $6s5d$, $6s6p$. Ca: $4s^2$, $4s3d$, $4s4p$. In each diagram the close-spaced triplet is the 3P or 3D multiplet; the isolated higher line is the singlet.

(e) Singlet–triplet ordering

Triplet space part is antisymmetric, electrons avoid each other; singlet symmetric, electrons overlap, larger Coulomb. Exchange splitting

$$E({}^1L) - E({}^3L) = 2K_{ns,n'\ell}, \quad K = \int d^3r_1 d^3r_2 \psi_{ns}^*(1) \psi_{n'\ell}^*(2) \frac{e^2}{4\pi\epsilon_0 r_{12}} \psi_{n'\ell}(1) \psi_{ns}(2) > 0. \quad (3)$$

(f) Interval rule

$H_{SO} = A \mathbf{L} \cdot \mathbf{S}$ in $|LSJM_J\rangle$ gives Landé's

$$E_J - E_{J-1} = A J. \quad (4)$$

Requires (i) pure LS, fine structure $\ll$ term splitting; (ii) one A per term (single-configuration). Hence *much better obeyed in Ca than Ba*: e.g. Ca $4s4p {}^3P$: $(E_2 - E_1)/(E_1 - E_0) \approx 1.96$ (cf. predicted 2); Ba $6s6p {}^3P$ ratio is significantly distorted.

(g) Allowed E1 transitions in Ba

Selection rules: parity flip ($\Delta\ell = \pm 1$ on one electron); $\Delta S = 0$; $\Delta L = 0, \pm 1$ ($0 \not\rightarrow 0$); $\Delta J = 0, \pm 1$ ($0 \not\rightarrow 0$). From $6s^2 {}^1S_0$ only $6s6p {}^1P_1$ is reachable ($s \rightarrow d$ forbidden by parity, triplets by ΔS). Within excited configurations:

$$\begin{aligned} \text{Resonance: } 6s^2 {}^1S_0 &\rightarrow 6s6p {}^1P_1, \\ {}^3D \rightarrow {}^3P : {}^3D_1 &\rightarrow {}^3P_{0,1,2}; {}^3D_2 \rightarrow {}^3P_{1,2}; {}^3D_3 \rightarrow {}^3P_2, \\ {}^1D \rightarrow {}^1P : {}^1D_2 &\rightarrow {}^1P_1. \end{aligned}$$

Strongest: $\boxed{6s^2 {}^1S_0 \leftrightarrow 6s6p {}^1P_1}$ resonance line: connects populated ground level, $\Delta S = 0$ (no spin-forbidden suppression), largest ω^3 in A_{21} and large $\langle 6s | r | 6p \rangle$.

Question 2 — Hydrogenic energy scales and positronium in a magnetic field

Problem. (a) H gross/fine/hyperfine scales; (b) Ps Bohr levels and $\langle r_{ep} \rangle$; (c)–(e) eigenstates, good QNs, energies and sketch for $H = A \mathbf{s}_e \cdot \mathbf{s}_p + 2\mu_B (\mathbf{s}_e - \mathbf{s}_p) \cdot \mathbf{B}$; (f) fine/hyperfine of Ps p -configurations.

(a) Energy scales

Gross (Rydberg):

$$E_{\text{gross}} \sim \frac{1}{2} m_e c^2 \alpha^2 \ (\approx 13.6 \text{ eV}). \quad (5)$$

Fine (relativistic, SO, Darwin), one extra $(v/c)^2 \sim \alpha^2$:

$$E_{\text{fs}} \sim m_e c^2 \alpha^4 \ (\approx 10^{-4} \text{ eV}). \quad (6)$$

Hyperfine: nuclear $\mu_p \sim \mu_N \propto 1/m_p$ vs. $\mu_B \propto 1/m_e$, so suppressed by m_e/m_p :

$$E_{\text{hfs}} \sim m_e c^2 \alpha^4 \frac{m_e}{m_p} \ (\approx 10^{-7} \text{ eV}). \quad (7)$$

SO is electron magnetic moment in motional $B \sim \alpha^2 m_e^2 c^2 / (e\hbar)$; hfs replaces one μ_B by μ_p , hence ratio $m_e/m_p \approx 1/1836$.

(b) Bohr Ps

Reduced mass $\mu = m_e/2$, so $E_n = -\mu c^2 \alpha^2 / (2n^2) = \frac{1}{2} E_n^{\text{H}}$:

$$|E_1^{\text{Ps}}| = \frac{1}{2} \times 13.6 \text{ eV} \approx 6.8 \text{ eV}. \quad (8)$$

Relative-coordinate Bohr radius $a_{\text{rel}} = \hbar / (\mu c \alpha) = 2a_0$, hence

$$\langle r_{ep} \rangle_{1s} \sim 2a_0. \quad (9)$$

(c) Eigenstate count and good quantum numbers

$$2 \times 2 = \boxed{4 \text{ eigenstates}}.$$

Low field ($A \gg 2\mu_B B$): hyperfine dominates; good QNs $(S_{\text{tot}}, M_{S_{\text{tot}}})$ with $S_{\text{tot}} = 0$ (para) or 1 (ortho).

High field ($2\mu_B B \gg A$): Zeeman dominates; field couples to $\mathbf{s}_e - \mathbf{s}_p$ so $\mathbf{S}_{\text{tot}}$ not conserved; good QNs (m_{s_e}, m_{s_p}) .

(d) Energies

Zero field. Square the total spin:

$$\mathbf{S}_{\text{tot}}^2 = (\mathbf{s}_e + \mathbf{s}_p)^2 = \mathbf{s}_e^2 + \mathbf{s}_p^2 + 2\mathbf{s}_e \cdot \mathbf{s}_p.$$

Solve for the dot product:

$$\mathbf{s}_e \cdot \mathbf{s}_p = \frac{1}{2} (\mathbf{S}_{\text{tot}}^2 - \mathbf{s}_e^2 - \mathbf{s}_p^2).$$

Take expectation in $|S_{\text{tot}}, M\rangle$ with $s_e = s_p = \frac{1}{2}$:

$$\langle \mathbf{s}_e \cdot \mathbf{s}_p \rangle = \frac{\hbar^2}{2} [S_{\text{tot}}(S_{\text{tot}} + 1) - \frac{3}{2}].$$

Evaluate the two cases:

$$\langle \mathbf{s}_e \cdot \mathbf{s}_p \rangle = \begin{cases} +\frac{1}{4}\hbar^2, & S_{\text{tot}} = 1 \\ -\frac{3}{4}\hbar^2, & S_{\text{tot}} = 0. \end{cases}$$

Multiply by A :

$$E_{S=1}^{(0)} = +\frac{1}{4} A \hbar^2 \ (\times 3), \quad E_{S=0}^{(0)} = -\frac{3}{4} A \hbar^2; \quad \text{splitting } A \hbar^2. \quad (10)$$

High field. Choose $\mathbf{B} = B\hat{\mathbf{z}}$ and the product basis $|m_{s_e}, m_{s_p}\rangle$. Expand the Zeeman term:

$$H_Z = 2\mu_B B (s_{ez} - s_{pz}).$$

Act on the product basis:

$$H_Z |m_{s_e}, m_{s_p}\rangle = 2\mu_B B \hbar (m_{s_e} - m_{s_p}) |m_{s_e}, m_{s_p}\rangle.$$

Decompose the hyperfine operator into z and ladder pieces:

$$\mathbf{s}_e \cdot \mathbf{s}_p = s_{ez}s_{pz} + \frac{1}{2}(s_{e+}s_{p-} + s_{e-}s_{p+}).$$

Drop off-diagonal ladder terms (suppressed by large Zeeman gap):

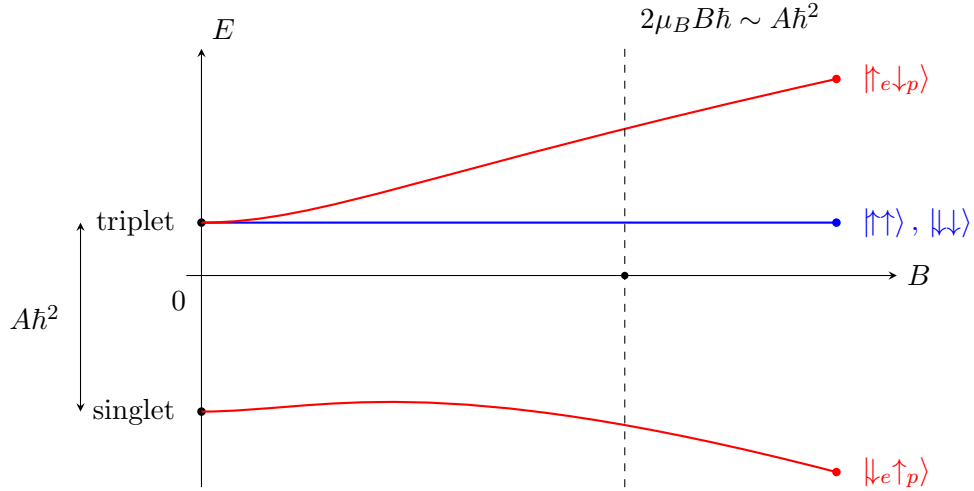
$$\langle \mathbf{s}_e \cdot \mathbf{s}_p \rangle \rightarrow \hbar^2 m_{s_e} m_{s_p}.$$

Add the two contributions for each product state:

$$\begin{aligned} \uparrow_e \uparrow_p &: E = +\frac{1}{4}A\hbar^2, \\ \downarrow_e \downarrow_p &: E = +\frac{1}{4}A\hbar^2, \\ \uparrow_e \downarrow_p &: E = -\frac{1}{4}A\hbar^2 + 2\mu_B B \hbar, \\ \downarrow_e \uparrow_p &: E = -\frac{1}{4}A\hbar^2 - 2\mu_B B \hbar. \end{aligned} \tag{11}$$

Stretched $\uparrow\uparrow, \downarrow\downarrow$ unaffected ($s_e - s_p$ has $m = 0$ on these).

(e) Sketch versus B



Crossover at $B^* = A\hbar/(2\mu_B)$; high-field slopes $\pm 2\mu_B \hbar$.

(f) Fine vs. hyperfine in Ps

In Ps the “nucleus” has $m_p \rightarrow m_e$, so the m_e/m_p suppression vanishes: *fine and hyperfine merge at* $\sim m_e c^2 \alpha^4 \sim 10^{-4}$ eV. For $\ell = 1$ Ps: 1P_1 (para), $^3P_{0,1,2}$ (ortho), with SO + spin-spin (contact + tensor) all of the same order. Additional Ps-only contribution: virtual e^+e^- annihilation into a virtual photon shifts $S_{\text{tot}} = 1$ levels at the same α^4 order. The fine/hyperfine separation loses meaning here.

Question 3 — Two-level Rabi oscillations and inhomogeneous broadening

Problem. Two-level atom in $\mathbf{E} = \mathcal{E}_0 \cos(\omega t) \hat{\mathbf{z}}$ for time T ; given

$$|c_2(T)|^2 = \frac{\Omega^2}{\Omega^2 + \delta^2} \sin^2\left(\frac{1}{2}\sqrt{\Omega^2 + \delta^2} T\right). \quad (12)$$

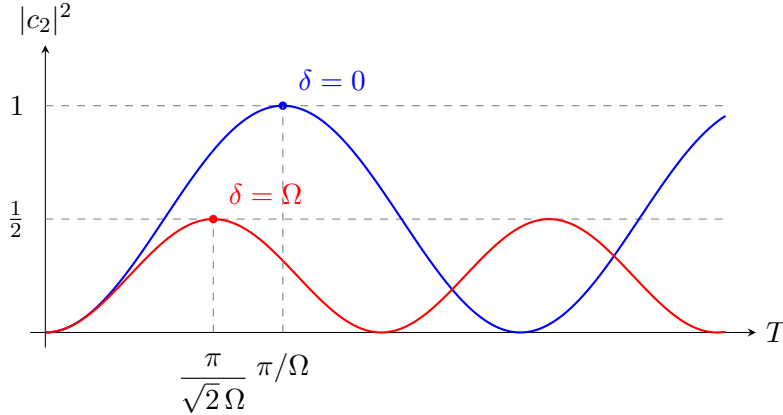
(a) Ω and δ

$$\boxed{\Omega = \frac{e\mathcal{E}_0}{\hbar} \langle 2 | \hat{z} | 1 \rangle \equiv \frac{\mathcal{E}_0 d_{12}}{\hbar}, \quad \delta = \omega - \omega_0, \quad \omega_0 = \frac{E_2 - E_1}{\hbar}.} \quad (13)$$

(b) Assumptions

(1) Two-level: other states off-resonance, $|\delta_{\text{other}}| \gg \Omega$. (2) Electric dipole: $ka \ll 1$, i.e. $\lambda \gg a_0$. (3) Semi-classical field (large photon number, no spontaneous emission). (4) No spontaneous decay, $T \ll \tau$. (5) RWA: drop counter-rotating $e^{\pm 2i\omega t}$, valid for $\Omega, |\delta| \ll \omega, \omega_0$. (6) Initial condition $c_1(0) = 1$, $c_2(0) = 0$.

(c) Sketches



$\delta = 0$: $|c_2|^2 = \sin^2(\Omega T/2)$, full inversion at $T = \pi/\Omega$. $\delta = \Omega$: amplitude $\frac{1}{2}$, period $2\pi/(\sqrt{2}\Omega)$, first peak at $T = \pi/(\sqrt{2}\Omega)$.

(d) Short-time limit

For $\sqrt{\Omega^2 + \delta^2} T \ll 1$, apply $\sin x \approx x$:

$$\sin^2\left(\frac{1}{2}\sqrt{\Omega^2 + \delta^2} T\right) \approx \frac{1}{4}(\Omega^2 + \delta^2) T^2.$$

Multiply by the prefactor:

$$|c_2(T)|^2 \approx \frac{\Omega^2}{\Omega^2 + \delta^2} \cdot \frac{1}{4}(\Omega^2 + \delta^2) T^2.$$

Cancel $(\Omega^2 + \delta^2)$:

$$|c_2(T)|^2 \approx \frac{(\Omega T)^2}{4}, \quad (14)$$

independent of δ : time–energy uncertainty $\sim 1/T$ exceeds δ , so all detunings within the pulse bandwidth are excited equally. Condition: $T \ll 1/\sqrt{\Omega^2 + \delta^2}$.

(e) **On-resonance π -pulse**

$$\boxed{T_\pi = \frac{\pi}{\Omega}.} \quad (15)$$

(f) **Ensemble average, $\Delta \ll \Omega$**

Substitute $T = T_\pi = \pi/\Omega$ into (12):

$$P_2(\delta) = \frac{\Omega^2}{\Omega^2 + \delta^2} \sin^2\left(\frac{\pi}{2} \sqrt{1 + \delta^2/\Omega^2}\right).$$

Binomial expand the prefactor:

$$\frac{\Omega^2}{\Omega^2 + \delta^2} = \frac{1}{1 + \delta^2/\Omega^2} \approx 1 - \frac{\delta^2}{\Omega^2} + \mathcal{O}(\delta^4/\Omega^4).$$

Expand the square root to first order:

$$\sqrt{1 + \delta^2/\Omega^2} \approx 1 + \frac{\delta^2}{2\Omega^2}.$$

The sine argument becomes

$$\frac{\pi}{2} \sqrt{1 + \delta^2/\Omega^2} \approx \frac{\pi}{2} + \frac{\pi\delta^2}{4\Omega^2}.$$

Apply the identity $\sin(\pi/2 + x) = \cos x$:

$$\sin^2\left(\frac{\pi}{2} + \frac{\pi\delta^2}{4\Omega^2}\right) = \cos^2\left(\frac{\pi\delta^2}{4\Omega^2}\right).$$

Cosine to second order:

$$\cos\left(\frac{\pi\delta^2}{4\Omega^2}\right) \approx 1 - \frac{1}{2}\left(\frac{\pi\delta^2}{4\Omega^2}\right)^2 = 1 + \mathcal{O}(\delta^4/\Omega^4).$$

Square it:

$$\cos^2\left(\frac{\pi\delta^2}{4\Omega^2}\right) = 1 + \mathcal{O}(\delta^4/\Omega^4).$$

Combine with the prefactor:

$$P_2(\delta) \approx \left(1 - \frac{\delta^2}{\Omega^2}\right)(1 + \mathcal{O}(\delta^4/\Omega^4)) = 1 - \frac{\delta^2}{\Omega^2}.$$

Take the ensemble mean over uniform $\delta \in [-\Delta/2, \Delta/2]$:

$$\langle \delta^2 \rangle = \frac{1}{\Delta} \int_{-\Delta/2}^{\Delta/2} \delta^2 d\delta = \frac{\Delta^2}{12}.$$

Insert into $\langle P_2 \rangle$:

$$\boxed{\langle P_2 \rangle = 1 - \frac{1}{12} \left(\frac{\Delta}{\Omega}\right)^2.} \quad (16)$$

Impose $\langle P_2 \rangle > 0.99$:

$$1 - \frac{1}{12} \left(\frac{\Delta}{\Omega}\right)^2 > 0.99.$$

Rearrange:

$$\frac{\Delta^2}{12\Omega^2} < 0.01.$$

Solve for Ω/Δ :

$$\left(\frac{\Omega}{\Delta}\right)^2 > \frac{1}{0.12} \approx 8.33.$$

$$\boxed{\Omega/\Delta \gtrsim 2.9.} \quad (17)$$

(g) Long-time limit

$\sqrt{\Omega^2 + \delta^2} T \gg 1$: $\sin^2$ averages to $\frac{1}{2}$. With $\Omega \gg \Delta$ prefactor $\rightarrow 1$. Hence

$$\boxed{\langle P_2 \rangle \rightarrow \frac{1}{2}.} \quad (18)$$

Dephasing across the ensemble ($\Delta T \gg 1$) destroys the coherent oscillation; “half-saturated” incoherent steady state.

Question 4 — Line broadening, saturation and Doppler-free spectroscopy

Problem. (a) Homogeneous vs. inhomogeneous; (b)–(d) rate equation, dI/dz in saturated form, identify κ, I_s ; (e)–(f) sketch ^{85}Rb cell ($L = 10\text{ cm}$, $N_T = 10^8\text{ cm}^{-3}$, $\lambda_0 = 780\text{ nm}$, $\tau = 27\text{ ns}$, $\sigma_{\text{peak}} = 3 \times 10^{-9}\text{ cm}^2$) at 1 mK and 10 K, then add saturating counter-pump.

(a) Homogeneous vs. inhomogeneous

Homogeneous: every atom has the same broadened line. Examples: natural (Lorentzian, FWHM $\Gamma = 1/\tau$); pressure/collisional. **Inhomogeneous:** different atoms have different transition frequencies; ensemble line is convolution of single-atom line with the distribution. Example: Doppler (Gaussian). Inhomogeneous broadening can be undone (saturation spectroscopy, photon echoes); homogeneous cannot.

(b) Rate equation

$$\frac{dN_2}{dt} = (N_1 - N_2) \sigma(\omega - \omega_0) \frac{I}{\hbar\omega} - \frac{N_2}{\tau}. \quad (19)$$

(c) Absorbed power and Beer-like equation

Spontaneous emission radiates into 4π , not the beam, so only net stimulated processes attenuate I :

$$\mathcal{P}_{\text{abs}} = (N_1 - N_2) \sigma(\omega - \omega_0) I \equiv \Delta N \sigma(\omega - \omega_0) I.$$

$$\boxed{\frac{dI}{dz} = -\Delta N \sigma(\omega - \omega_0) I.} \quad (20)$$

(d) Steady-state imbalance and saturation

Set $dN_2/dt = 0$ in (19):

$$(N_1 - N_2) \sigma \frac{I}{\hbar\omega} = \frac{N_2}{\tau}.$$

Define the dimensionless intensity $s = \sigma I \tau / (\hbar\omega)$:

$$N_2 = (N_1 - N_2) s.$$

Expand and collect N_2 :

$$N_2(1 + s) = N_1 s.$$

Solve for N_2 in terms of N_1 :

$$N_2 = \frac{s}{1 + s} N_1.$$

Use total population $N_1 + N_2 = N_T$ with $N_1 = N_T - N_2$:

$$N_2(1 + s) = (N_T - N_2) s.$$

Collect N_2 again:

$$N_2(1 + 2s) = N_T s.$$

Solve for the populations:

$$N_1 = \frac{1 + s}{1 + 2s} N_T, \quad N_2 = \frac{s}{1 + 2s} N_T.$$

Form the imbalance $\Delta N = N_1 - N_2$:

$$\Delta N = \frac{N_T}{1 + 2s}.$$

Substitute into (20):

$$\frac{dI}{dz} = -\frac{N_T \sigma I}{1 + 2s}.$$

Identify $2s = I/I_s$ with $I_s = \hbar\omega/(2\sigma\tau)$:

$$\frac{dI}{dz} = -\frac{N_T \sigma I}{1 + I/I_s}, \quad \boxed{\kappa(\omega) = N_T \sigma(\omega - \omega_0), \quad I_s(\omega) = \frac{\hbar\omega}{2 \sigma(\omega - \omega_0) \tau}.} \quad (21)$$

At $I = I_s$, ΔN halves and absorption halves; factor 2 because each absorption depletes N_1 and adds to N_2 .

(e) Weak-beam profiles

For $I \ll I_s$, $1 + I/I_s \rightarrow 1$, so the saturated equation reduces to the linear Beer form. Separate variables:

$$\frac{dI}{I} = -N_T \sigma dz.$$

Integrate from 0 to z :

$$\ln I(z) - \ln I_0 = -N_T \sigma z.$$

Exponentiate:

$$I(z) = I_0 \exp[-N_T \sigma z],$$

so $I(L) = I_0 \exp[-N_T \sigma L]$.

Natural FWHM: $\Delta\nu_{\text{nat}} = 1/(2\pi\tau) \approx 5.9 \text{ MHz}$.

Doppler FWHM: with $m = 85 m_u = 1.41 \times 10^{-25} \text{ kg}$, $\lambda_0 = 780 \text{ nm}$,

$$\Delta\nu_D = \frac{1}{\lambda_0} \sqrt{\frac{8k_B T \ln 2}{m}}.$$

Plug in $T = 10 \text{ K}$:

$$\Delta\nu_D(10 \text{ K}) = \frac{1}{780 \times 10^{-9} \text{ m}} \sqrt{\frac{8 \times 1.38 \times 10^{-23} \text{ J/K} \times 10 \text{ K} \times \ln 2}{1.41 \times 10^{-25} \text{ kg}}}.$$

Reduce the radical:

$$\Delta\nu_D(10 \text{ K}) \approx \frac{73.7 \text{ m/s}}{780 \times 10^{-9} \text{ m}} \approx 94 \text{ MHz}.$$

Scale by $\sqrt{T}$ (factor 10^{-2}) to get $T = 1 \text{ mK}$:

$$\Delta\nu_D(1 \text{ mK}) \approx 0.94 \text{ MHz}.$$

At 1 mK: Doppler (0.94 MHz) $\ll$ natural (5.9 MHz), so the line is natural-dominated Lorentzian, FWHM $\approx 5.9 \text{ MHz}$, $\sigma \approx \sigma_{\text{peak}}$. Compute optical depth:

$$N_T \sigma L \approx 10^8 \text{ cm}^{-3} \times 3 \times 10^{-9} \text{ cm}^2 \times 10 \text{ cm} = 3.$$

On-resonance transmission:

$$e^{-3} \approx 5\%.$$

At 10 K: Doppler (94 MHz) $\gg$ natural (5.9 MHz), so the line is Doppler-dominated Gaussian, FWHM $\approx 94 \text{ MHz}$. Peak σ reduced by $\Gamma_{\text{nat}}/\Delta\omega_D$:

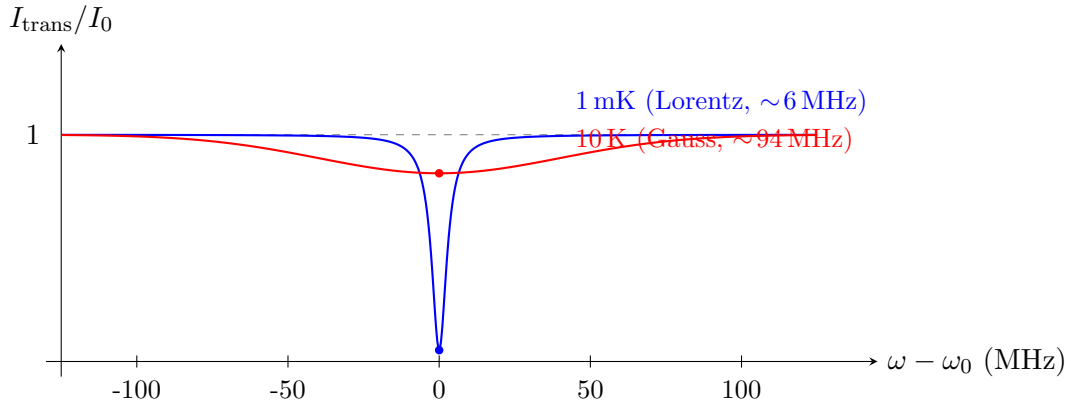
$$\sigma_{\text{peak}}^{\text{eff}}/\sigma_{\text{peak}} \approx 5.9 \text{ MHz}/94 \text{ MHz} \approx 0.063.$$

Compute optical depth:

$$N_T \sigma L \approx 3 \times 0.063 \approx 0.19.$$

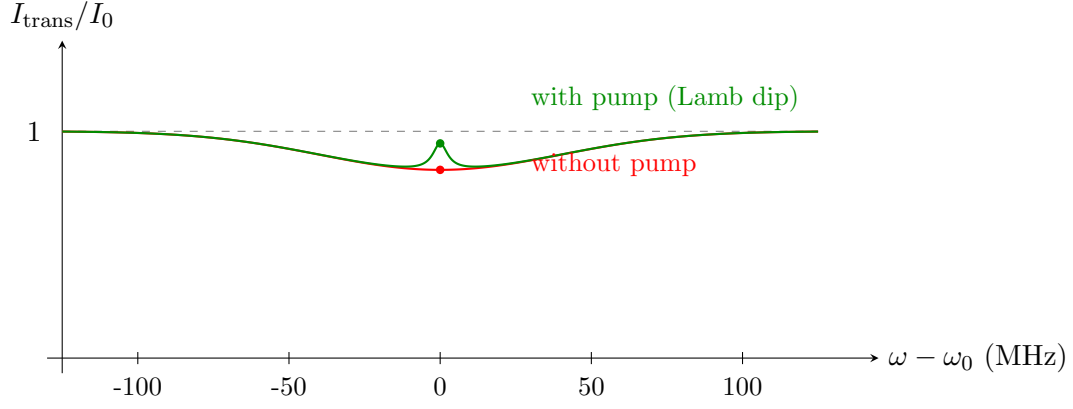
On-resonance transmission:

$$e^{-0.19} \approx 83\%.$$



(f) **Doppler-free saturation at 10 K**

Counter-propagating pump at ω_0 , $I_p \sim I_s$. Atom of velocity v_z sees probe at $\omega(1 + v_z/c)$, pump at $\omega_0(1 - v_z/c)$. Pump saturates only $v_z = 0$ atoms; probe is on resonance with that class only at $\omega = \omega_0$. A narrow **Lamb dip** (here, a peak in transmission) of width $\sim \Gamma_{\text{nat}} \approx 6$ MHz appears at line centre on the broad Doppler dip:



This is **Doppler-free saturation spectroscopy**: removes inhomogeneous broadening at the cost of a strong pump.